

Lecture Notes

on Assorted Topics in Category Theory

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0 Welcome

0.1 Introduction

This document contains an assortment of lecture notes prepared initially for use in lectures of 2–3 hours, focused at a listener with a decent amount of mathematical maturity and some experience with category theory, algebra, topology, and so on. There is no overarching goal, other than to cover interesting topics in and around category theory, homological algebra, geometry, higher category theory, and higher algebra.

Very often, the notes have been prepared without too much care, and so one should expect mistakes, occasional hand-waving, and sub-optimal approaches. Hopefully, all the essential ideas remain correct. On a related note, while there has been some effort put into making the exposition clear, we are sometimes rather terse as it makes it easier to use the notes for lectures, where the exposition may be expanded upon in any case.

References are not always carefully tracked. If in doubt, assume no originality.

0.2 Foundations

In this preliminary section, we spell out the general foundational framework we adopt in these notes. Since foundations will never play a *huge* role in what we do, or at least it won't affect how we do things generally, we will not be too careful. However, the crux of it is as follows: we adopt your favourite common axiomatics for set theory, such as ZFC, along with Grothendieck's universe axiom. A (Grothendieck) universe is essentially a model for ZF set theory within ZFC set theory itself, so it is a set wherein one can do all basic expected operations (and contains $\mathbb{N}$). The universe axiom postulates that every set is contained in a universe. In particular, we may pick some universe $\mathbb{U}_1 \ni \emptyset$, the elements of which we call the *small sets*. Iterating this, the axiom also implies the existence of a hierarchy of universes

$$\mathbb{U}_1 \in \mathbb{U}_2 \in \mathbb{U}_3 \in \dots \in \mathbb{U}_n \in \dots$$

and we call elements of $\mathbb{U}_2$ *large*, elements of $\mathbb{U}_3$ *very large*, and so on. If we have picked some universe $\mathbb{U}$, we may also refer to a set X as $\mathbb{U}$ -small to mean that $X \in \mathbb{U}$, or that X is in bijection with a set in $\mathbb{U}$. Note that we will essentially never need to use this nor think about it in any detail. To be noted as well is that in this framework, the notion of a *class* is subsumed by the notion of a set, in the sense that we may think of elements of $\mathbb{U}_2 \setminus \mathbb{U}_1$ as analogous to “proper classes.” If one wishes, type theory can play a similar role, where one demands a hierarchy of type universes instead.

0.3 Planned contents

Some planned contents already have a rough lecture section assigned to them, but some others are a lot looser and so are hard to pin down enough for that to be reasonable, and may span many lectures. Here is a list to which I aspire.

- (1) The theory of $(\infty, 1)$ -categories, blending the standard approaches (say, [Lur09] and [Cis19]) with the synthetic approach of [RV22].
- (2) Sheaves on sites, probably following [KS06]. Relating sheaves between sites will be a focus. Maybe something from [Lur09, §6.2.2] and [Lur18, §1.3.1] to include ∞ -categorical aspects.
- (3) Approximable triangulated categories.
- (4) Perverse sheaves, as an application of the gluing of t-structures.
- (5) Brown representability.

- (6) Topics surrounding model categories: Bousfield localizations, stable model categories, algebraic small object argument?
- (7) Operads, due to their importance in understanding e.g. E_n -rings and their modules.
- (8) Huber's adic spaces, as a "reasonably elementary" formalism for rigid analytic geometry.
- (9) Condensed sets, with a focus on eventually building up to condensed analytic geometry (via Clausen & Scholze's analytic stacks).
- (10) K -theory, particularly of stable ∞ -categories or Waldhausen ∞ -categories. Perhaps recent research extending the K -theory machine to certain large categories (dualizable ones), à la Efimov.
- (11) Waldhausen categories, i.e. categories with cofibrations and weak equivalences.
- (12) "Lower" higher category theory, so 2-categories and bicategories. (Virtual) double categories, for their applications to ∞ -cosmoi.
- (13) Topoi and ∞ -topoi.
- (14) The theorem of Hoshino–Kato–Miyachi on t-structures generated by silting objects. Also requires the computation of Ext^1 in the heart of a t-structure in terms of the Hom-sets in the ambient triangulated category.
- (15) Witt vectors.
- (16) Tensor triangular geometry.

18 Graded modules & algebras

In areas incorporating homological algebra, including homological algebra itself, one is commonly concerned with chain complexes of modules over a ring. However, often these chain complexes admit a bit more structure than merely being chain complexes: they can in many cases admit a kind of multiplication. Forgetting the differential, one has merely *graded* modules.

18.1 Graded objects

Let us fix a commutative ring $\mathbb{k}$. Here, we use $\mathbb{k}$ to denote a *ring* instead of a *field* for historical reasons. Commonly, a $(\mathbb{Z})$ -graded $\mathbb{k}$ -module is said to be a $\mathbb{k}$ -module M together with a decomposition

$$M \cong \coprod_{i \in \mathbb{Z}} M^i$$

of M as a $\mathbb{Z}$ -indexed coproduct of $\mathbb{k}$ -modules—the *grading* on M . It is not hard to see that this datum is really equivalent to M , and in particular, it makes sense to make the following definition.

Definition 18.1. Let $\mathcal{C}$ be a category. The category of $\mathbb{Z}$ -graded objects in $\mathcal{C}$ is the functor category

$$\text{Fun}(\mathbb{Z}, \mathcal{C}) \cong \prod_{\mathbb{Z}} \mathcal{C},$$

where $\mathbb{Z}$ is regarded as a discrete category. In particular, we obtain the category $\mathbf{gMod}_{\mathbb{k}}$ of *graded $\mathbb{k}$ -modules*

$$\mathbf{gMod}_{\mathbb{k}} := \prod_{\mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}.$$

For $x \in \text{Fun}(\mathbb{Z}, \mathcal{C})$, we call x^i the *degree i piece* of x . For $M \in \mathbf{gMod}_{\mathbb{k}}$, we say that an element of M^i is *homogeneous of degree i* .

Remark 18.2. We will essentially always be working with $\mathbb{Z}$ -graded objects, so henceforth we will drop bothering to specify that.

Remark 18.3. One could have defined graded $\mathbb{k}$ -modules as being pairs $(M, (M^i)_{i \in \mathbb{Z}})$, with morphisms being the $\mathbb{k}$ -linear maps which preserve the grading. It is easily seen that this results in an equivalent category.

Terminology 18.4. Let $M, N \in \mathbf{gMod}_{\mathbb{k}}$. A morphism $M \rightarrow N$ in $\mathbf{gMod}_{\mathbb{k}}$ is said to be a *graded morphism of degree 0*, or otherwise a *strict morphism*.

Let us translate some of the terminology standard to graded objects to our setting. In particular, let us detail how one encodes morphisms of non-zero degree in this formalism.

Definition 18.5. Let $\mathcal{C}$ be a category. The automorphism $k \mapsto k+1$ of $\mathbb{Z}$ induces the automorphism

$$(1) : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \text{Fun}(\mathbb{Z}, \mathcal{C}), \quad x = (x^i)_{i \in \mathbb{Z}} \mapsto (x^{i+1})_{i \in \mathbb{Z}} =: x(1).$$

of categories of graded objects in $\mathcal{C}$. We denote by (-1) the inverse of this functor, and thus produce functors (i) for all $i \in \mathbb{Z}$ by repeated application of either (1) or (-1) .

Terminology 18.6. Let $M, N \in \mathbf{gMod}_{\mathbb{k}}$, and let $i \in \mathbb{Z}$. A *morphism of degree i* from M to N is a strict morphism $M \rightarrow N(i)$.

Remark 18.7. This definition makes sense: a morphism of degree i should send a degree n homogeneous element of M to a degree $n + i$ homogeneous element of N , and indeed, the above definition yields

$$M^n \rightarrow N^{n+i}.$$

Given a morphism $M \rightarrow N(i)$ and a morphism $N \rightarrow L(j)$, one can “compose” these by forming

$$M \rightarrow N(i) \rightarrow L(i+j),$$

thus obtaining a morphism from M to L of degree $i+j$.

Remark 18.8. Note that since (i) is an automorphism for all $i \in \mathbb{Z}$, we have that

$$\mathbf{gMod}_{\mathbb{k}}(M, N) \cong \mathbf{gMod}_{\mathbb{k}}(M(i), N(i)).$$

In particular, we get no value from considering maps of the form $M(i) \rightarrow N(j)$, as we can always assume $i = 0$.

18.2 Tensor products and Hom in the graded world

We will now turn to discussing a little bit of actual algebra, in contrast to the above entirely pure definitions. We will define the *tensor product* of two graded modules in such a way that it is a graded module itself, and similarly produce a graded version of Hom. For the purposes of the former, it is useful to introduce *bigraded* modules. These are just modules graded over $\mathbb{Z} \times \mathbb{Z}$, defined just as one does for $\mathbb{Z}$.

Definition 18.9. Let $\mathcal{C}$ be a category. The category of *bigraded objects* in $\mathcal{C}$ is the functor category

$$\mathrm{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{C}) \cong \prod_{\mathbb{Z} \times \mathbb{Z}} \mathcal{C}.$$

In particular, we obtain the category

$$\mathbf{bgMod}_{\mathbb{k}} := \prod_{\mathbb{Z} \times \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}.$$

The reason we introduce this is because given two graded modules M, N , the most natural way to form a tensor product is to form the *bigraded* module

$$(M^i \otimes_{\mathbb{k}} N^j), \quad (i, j) \in \mathbb{Z} \times \mathbb{Z}.$$

A problem with this, of course, is that we are interested in the world of *graded* modules, not *bigraded* modules. The solution is:

Definition 18.10. Let $\mathcal{C}$ be a category. We define the functor

$$\mathrm{Tot} : \mathrm{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{C}) \rightarrow \mathrm{Fun}(\mathbb{Z}, \mathcal{C}), \quad (x^{i,j})_{(i,j) \in \mathbb{Z} \times \mathbb{Z}} \mapsto \left(\prod_{i+j=n} x^{i,j} \right)_{n \in \mathbb{Z}}.$$

The way to visualize this functor is by picturing the bigraded object as lying in a grid, then taking the coproduct of the diagonal lines:

$$\begin{array}{cccc}
 x^{-1,2} & x^{0,2} & x^{1,2} & x^{2,2} \\
 \swarrow & \swarrow & \swarrow & \swarrow \\
 x^{-1,1} & x^{0,1} & x^{1,1} & x^{2,1} \\
 \swarrow & \swarrow & \swarrow & \swarrow \\
 x^{-1,0} & x^{0,0} & x^{1,0} & x^{2,0} \\
 \swarrow & \swarrow & \swarrow & \swarrow \\
 x^{-1,-1} & x^{0,-1} & x^{1,-1} & x^{2,-1}
 \end{array}$$

Proposition 18.11. Let $\mathcal{C}$ be a category admitting countable coproducts. Consider the functor

$$G : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \text{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{C}), \quad (x^i)_{i \in \mathbb{Z}} \mapsto (x^{i+j})_{(i,j) \in \mathbb{Z} \times \mathbb{Z}}.$$

Then $\text{Tot} \vdash G$.

Proof. For $x \in \text{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{C})$ and $y \in \text{Fun}(\mathbb{Z}, \mathcal{C})$, we have

$$\begin{aligned} \text{Hom}(\text{Tot}(x), y) &= \prod_{n \in \mathbb{Z}} \text{Hom}(\text{Tot}(x)^n, y^n) \\ &= \prod_{n \in \mathbb{Z}} \text{Hom}\left(\coprod_{i+j=n} x^{i,j}, y^n\right) \\ &\cong \prod_{n \in \mathbb{Z}} \prod_{i+j=n} \text{Hom}(x^{i,j}, y^n) \\ &= \prod_{n \in \mathbb{Z}} \prod_{i+j=n} \text{Hom}(x^{i,j}, G(y)^{i,j}) \\ &= \prod_{(i,j) \in \mathbb{Z} \times \mathbb{Z}} \text{Hom}(x^{i,j}, G(y)^{i,j}) = \text{Hom}(x, G(y)) \end{aligned}$$

as desired. ■

Definition 18.12. Let $M, N \in \mathbf{gMod}_{\mathbb{k}}$. Temporarily, let us denote the bigraded $\mathbb{k}$ -module $(M^i \otimes_{\mathbb{k}} N^j)_{(i,j) \in \mathbb{Z} \times \mathbb{Z}}$ by $M \otimes_{\mathbb{k}}^b N$. We define the graded modules

$$\begin{aligned} M \otimes_{\mathbb{k}} N &:= \text{Tot}(M \otimes_{\mathbb{k}}^b N), \\ \mathbf{gMod}_{\mathbb{k}}(M, N) &:= (\mathbf{gMod}_{\mathbb{k}}(M, N(i)))_{i \in \mathbb{Z}}. \end{aligned}$$

We will also write $\underline{\text{Hom}}(M, N) := \mathbf{gMod}_{\mathbb{k}}(M, N)$ if the context makes it clear what this means. These clearly organize into functors.

Proposition 18.13. Let $M, N, L \in \mathbf{gMod}_{\mathbb{k}}$. There is a natural isomorphism

$$\mathbf{gMod}_{\mathbb{k}}(M \otimes_{\mathbb{k}} N, L) \cong \mathbf{gMod}_{\mathbb{k}}(M, \mathbf{gMod}_{\mathbb{k}}(N, L)).$$

In particular, we have an adjunction $- \otimes_{\mathbb{k}} N \vdash \mathbf{gMod}_{\mathbb{k}}(N, -)$.

Proof. Recall the adjunction $\text{Tot} \vdash G$ from Proposition 18.11. Then

$$\begin{aligned} \mathbf{gMod}_{\mathbb{k}}(M \otimes_{\mathbb{k}} N, L) &= \mathbf{gMod}_{\mathbb{k}}(\text{Tot}(M \otimes_{\mathbb{k}}^b N), L) \\ &\cong \mathbf{bgMod}_{\mathbb{k}}(M \otimes_{\mathbb{k}}^b N, G(L)) \\ &= \prod_{(i,j) \in \mathbb{Z} \times \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(M^i \otimes_{\mathbb{k}} N^j, L^{i+j}) \\ &\cong \prod_{(i,j) \in \mathbb{Z} \times \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(M^i, \mathbf{Mod}_{\mathbb{k}}(N^j, L^{i+j})) \\ &= \prod_{i \in \mathbb{Z}} \prod_{j \in \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(M^i, \mathbf{Mod}_{\mathbb{k}}(N^j, L^{i+j})) \\ &\cong \prod_{i \in \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(M^i, \prod_{j \in \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(N^j, L^{i+j})) \\ &= \prod_{i \in \mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}(M^i, \mathbf{gMod}_{\mathbb{k}}(N, L(i))) = \mathbf{gMod}_{\mathbb{k}}(M, \mathbf{gMod}_{\mathbb{k}}(N, L)) \end{aligned}$$

as desired. ■

Remark 18.14. Note that the foregoing proof and definition goes through in the generality of a closed symmetric monoidal Abelian category. That is, an Abelian category $\mathcal{A}$ admitting a symmetric monoidal structure $\otimes$ such that $- \otimes x$ has a right adjoint for all $x \in \mathcal{A}$.

Remark 18.15. It is perhaps useful, for purposes of intuition, to observe that the elements homogeneous of degree i in $\underline{\mathbf{gMod}}_{\mathbb{k}}(M, N)$ are exactly the degree i maps from M to N .

Remark 18.16. By Proposition 18.11, to describe a map $f: M \otimes_{\mathbb{k}} N \rightarrow L$, it suffices to say how f acts on elementary tensors

$$x \otimes y, \quad x \in M^i, y \in N^j,$$

where one must ensure that $f(x \otimes y) \in L^{i+j}$.

Remark 18.17. The isomorphism $\mathbf{gMod}_{\mathbb{k}}(M \otimes_{\mathbb{k}} N, L) \cong \mathbf{gMod}_{\mathbb{k}}(M, \mathbf{gMod}_{\mathbb{k}}(N, L))$ is given explicitly by *currying*. That is, given a map $f: M \otimes_{\mathbb{k}} N \rightarrow L$, the corresponding map $M \rightarrow \mathbf{gMod}_{\mathbb{k}}(N, L)$ is defined by the formula

$$x \mapsto f(x \otimes -).$$

Note that if $x \in M^i$ and $y \in N^j$, then $x \otimes y \in (M \otimes_{\mathbb{k}} N)^{i+j}$, so $f(x \otimes y) \in L^{i+j}$, so the morphism $f(x \otimes -)$ does indeed have the expected type $N \rightarrow L(i)$.

The tensor product $- \otimes_{\mathbb{k}} - : \mathbf{gMod}_{\mathbb{k}} \times \mathbf{gMod}_{\mathbb{k}} \rightarrow \mathbf{gMod}_{\mathbb{k}}$ endows $\mathbf{gMod}_{\mathbb{k}}$ with the structure of a (closed) symmetric monoidal category. In particular, it is clear that one has natural isomorphisms $M \otimes_{\mathbb{k}} \mathbb{k} \cong M$, and that one has associativity isomorphisms

$$(M \otimes_{\mathbb{k}} N) \otimes_{\mathbb{k}} L \cong M \otimes_{\mathbb{k}} (N \otimes_{\mathbb{k}} L).$$

It is maybe a *little* less clear that these satisfy the various coherences required of a monoidal category, but the idea is that they are inherited from the ones on $\mathbf{Mod}_{\mathbb{k}}$. This justifies the assertion that we have a *monoidal* structure.

There is a slight complication in how we make the monoidal structure *symmetric*. In particular, the natural isomorphisms

$$\tau_{M,N}: M \otimes_{\mathbb{k}} N \xrightarrow{\sim} N \otimes_{\mathbb{k}} M$$

are chosen to act by the *Koszul sign rule*,

$$\tau(x \otimes y) = (-1)^{ij} y \otimes x, \quad x \in M^i, y \in N^j.$$

This sign convention is the fundamental reason that graded commutativity is different from ordinary commutativity. The reason one is interested in this form of commutativity is that one would like objects like the exterior algebra $\bigwedge^{\bullet} M$ of a module M to be examples of *graded (commutative) algebras*, since these arise often in e.g. algebraic geometry and topology. These exterior algebras satisfy $x \wedge y = (-1)^{ij} y \wedge x$.

Exercise 43. Show that there is a natural isomorphism

$$\mathbf{gMod}_{\mathbb{k}}(M \otimes N, L) \cong \mathbf{gMod}_{\mathbb{k}}(M, \mathbf{gMod}_{\mathbb{k}}(N, L)).$$

Hint: this is purely formal, and holds in any closed monoidal category.

Exercise 44. Check that $\mathbf{gMod}_{\mathbb{k}}$ is an Abelian category.

18.3 Tensor products of morphisms

We finish our present discussion of the tensor product by explaining how the Koszul sign rule affects *evaluation* of tensor products of morphisms. The adjunction from Proposition 18.13 induces a counit, and it is the *evaluation map*

$$\mathrm{ev}: \mathbf{gMod}_{\mathbb{k}}(M, N) \otimes_{\mathbb{k}} M \rightarrow N, \quad \mathrm{ev}(f \otimes x) = f(x).$$

Note that the formula on the right is slightly more subtle than it may first appear: the evaluation map is a map of *graded modules*, and so applies to morphisms of non-zero degree as well. More precisely, if $f \in \underline{\mathbf{gMod}}_{\mathbf{k}}(M, N)^i$ and $x \in M^j$, then $\text{ev}(f \otimes x) = f(x) \in N^{i+j}$, as expected, since f is the same as a degree zero map $M \rightarrow N(i)$.

When we have two strict morphisms $f: M \rightarrow L$, $g: N \rightarrow K$, the morphism $f \otimes g$ we obtain is simply given by

$$(f \otimes g)(x \otimes y) = f(x) \otimes g(y).$$

However, what we would like is a way to take tensor products of morphisms of non-zero degree—this does not come for free. Sure, we can regard a degree i morphism $M \rightarrow L$ as a strict morphism $M \rightarrow L(i)$ and take the tensor product that way, but this produces something of the wrong type (at least visually). What we want is a morphism

$$\underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} \underline{\mathbf{gMod}}_{\mathbf{k}}(N, K) \rightarrow \underline{\mathbf{gMod}}_{\mathbf{k}}(M \otimes_{\mathbf{k}} N, L \otimes_{\mathbf{k}} K)$$

taking the “formal” tensor product of two arbitrary degree maps to its “realization” as an actual morphism. Furthermore, we want this to be suitably compatible with the evaluation; to be more precise, using the associativity and symmetry isomorphisms together with the evaluation, we can produce the composite

$$\begin{aligned} & (\underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} \underline{\mathbf{gMod}}_{\mathbf{k}}(N, K)) \otimes_{\mathbf{k}} (M \otimes_{\mathbf{k}} N) \\ & \quad \downarrow \cong \\ & \underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} (\underline{\mathbf{gMod}}_{\mathbf{k}}(N, K) \otimes_{\mathbf{k}} M) \otimes_{\mathbf{k}} N \\ & \quad \downarrow 1 \otimes \tau \otimes 1 \cong \\ & \underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} (M \otimes_{\mathbf{k}} \underline{\mathbf{gMod}}_{\mathbf{k}}(N, K)) \otimes_{\mathbf{k}} N \\ & \quad \downarrow \cong \\ & (\underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} M) \otimes_{\mathbf{k}} (\underline{\mathbf{gMod}}_{\mathbf{k}}(N, K) \otimes_{\mathbf{k}} N) \\ & \quad \downarrow \text{ev} \otimes \text{ev} \\ & L \otimes_{\mathbf{k}} K \end{aligned}$$

and we would like for this to be the map which transposes, via currying, to the “realization” morphism

$$\underline{\mathbf{gMod}}_{\mathbf{k}}(M, L) \otimes_{\mathbf{k}} \underline{\mathbf{gMod}}_{\mathbf{k}}(N, K) \rightarrow \underline{\mathbf{gMod}}_{\mathbf{k}}(M \otimes_{\mathbf{k}} N, L \otimes_{\mathbf{k}} K).$$

In the end, this means that we are forced to make the following definition:

Definition 18.18. Let $f \in \underline{\mathbf{gMod}}_{\mathbf{k}}(M, L)$ and let $g \in \underline{\mathbf{gMod}}_{\mathbf{k}}(N, K)^i$. We define their tensor product $f \otimes g$ to be the degree $\deg(f) + i$ morphism $M \otimes_{\mathbf{k}} N \rightarrow L \otimes_{\mathbf{k}} K$ given by

$$(f \otimes g)(x \otimes y) = (-1)^{ij} f(x) \otimes g(y), \quad x \in M^j.$$

Remark 18.19. The sign depending on the degrees of g and x appearing in the definition comes directly from the Koszul sign rule, since we apply the isomorphism τ in the above composite.

18.4 Graded algebras

In the context of any symmetric monoidal category, one can define *monoid* objects. Applying that to our situation, we end up with

Definition 18.20. A *graded $\mathbb{k}$ -algebra* is a pair (A, μ) of a graded $\mathbb{k}$ -module A and a multiplication map

$$\mu: A \otimes_{\mathbb{k}} A \rightarrow A$$

which is associative and unital. That is, there is an element $1 = 1_A \in A^0$ such that for all $x \in A^i$, we have $\mu(1 \otimes x) = x = \mu(x \otimes 1)$, and one should have $\mu(\mu(x \otimes y) \otimes z) = \mu(x \otimes \mu(y \otimes z))$. We write $xy := \mu(x \otimes y)$.

Remark 18.21. As before, it is helpful to note that a map $A \otimes_{\mathbb{k}} A \rightarrow A$ consists of the data of maps

$$A^i \otimes_{\mathbb{k}} A^j \rightarrow A^{i+j}, \quad \forall i, j \in \mathbb{Z}.$$

The conditions we require on μ are just such that

$$1 \cdot x = x = x \cdot 1, \quad (xy)z = x(yz).$$

That is, the expected axioms.

Definition 18.22. We say that a graded algebra A is (*graded*) *commutative* if the diagram

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\mu} & A \\ \cong \downarrow \tau & & \parallel \\ A \otimes A & \xrightarrow{\mu} & A \end{array}$$

commutes. Concretely, this means that

$$xy = (-1)^{ij}yx, \quad x \in A^i, y \in A^j.$$

Remark 18.23. Note that the Koszul sign rule means that commutativity, in characteristic not equal to 2, necessarily means that

$$y^2 = 0$$

whenever y is homogeneous of odd degree. Indeed,

$$y^2 = (-1)^{\deg(y)\deg(y)}y^2 = -y^2 \implies 2y^2 = 0.$$

For this reason, one often *requires* that this holds as part of the definition in the even characteristic case. We will not make this distinction here, because we will often implicitly assume we are not working in a context where there will be problems.

Example 18.24. We now make precise the example from earlier. Let M be a $\mathbb{k}$ -module. We define $M \wedge_{\mathbb{k}} M$ to be the $\mathbb{k}$ -module quotient

$$M \wedge_{\mathbb{k}} M := (M \otimes_{\mathbb{k}} M) / (x \otimes y + y \otimes x)_{x, y \in M}.$$

We denote the image of $x \otimes y$ in $M \wedge_{\mathbb{k}} M$ by $x \wedge y$. The definition is such that $x \wedge y = -y \wedge x$.

Now, for all $k \geq 0$, we define

$$\bigwedge^k M := \underbrace{M \wedge_{\mathbb{k}} \cdots \wedge_{\mathbb{k}} M}_{k \text{ copies}}$$

and set $\bigwedge^0 M := \mathbb{k}$. We then have a graded $\mathbb{k}$ -module defined by

$$\left(\bigwedge M\right)^i = \begin{cases} \bigwedge^i M & \text{if } i \geq 0, \\ 0 & \text{if } i \leq 0. \end{cases}$$

This graded module can be endowed with the structure of a graded $\mathbb{k}$ -algebra, which is furthermore commutative. In particular, given elementary generators $x = x_1 \wedge \cdots \wedge x_i$ and $y = y_1 \wedge \cdots \wedge y_j$, one defines

$$x \wedge y = x_1 \wedge \cdots \wedge x_i \wedge y_1 \wedge \cdots \wedge y_j$$

and one easily sees that this extends to maps

$$\wedge : \left(\bigwedge M\right)^i \otimes_{\mathbb{k}} \left(\bigwedge M\right)^j \rightarrow \left(\bigwedge M\right)^{i+j}$$

and thus to yields a map

$$\bigwedge M \otimes_{\mathbb{k}} \bigwedge M \rightarrow \bigwedge M.$$

It is left as an exercise to check that this is associative and unital. That it is graded commutative comes from the definition of the wedge.

Example 18.25. Standard examples of commutative $\mathbb{k}$ -algebras are polynomial rings over $\mathbb{k}$, i.e. the *free* commutative algebras over $\mathbb{k}$. In the graded world, there are similar equivalents, but with the grading comes the freedom to choose what *cohomological degree*—i.e. what degree in the grading—the generators have.

Let us first consider the case of a single generator x , yielding a commutative graded $\mathbb{k}$ -algebra $\mathbb{k}[x]$. If we decide that x is to be of degree zero, then this is just the ordinary polynomial ring in one variable over $\mathbb{k}$. Otherwise, the description of $\mathbb{k}[x]$ splits into two cases depending on whether the degree ℓ of x is even or odd, and positive or negative. Since the difference between negative and positive isn't very interesting, let us consider the case when $\ell < 0$.

If ℓ is even, then $\mathbb{k}[x]$ as a graded $\mathbb{k}$ -module looks like this:

$$\cdots \quad \mathbb{k} \cdot x^k \quad 0 \quad \cdots \quad \mathbb{k} \cdot x \quad 0 \quad \cdots \quad \mathbb{k} \quad 0 \quad \cdots$$

where a copy of $\mathbb{k}$ is placed in positions $k\ell$ for all $k \leq 0$, and everything else is zero. The multiplication is such that $x^i x^j = x^{i+j}$. In particular, $\mathbb{k}[x]$ has an infinite number of copies of $\mathbb{k}$ in it.

If ℓ is odd, then $\mathbb{k}[x]$ is considerably simpler. It has the following form:

$$\cdots \quad 0 \quad \mathbb{k} \cdot x \quad 0 \quad \cdots \quad 0 \quad \mathbb{k} \quad 0 \quad \cdots$$

where there is a copy of $\mathbb{k}$ placed in degree 0 and degree ℓ . This is because we necessarily have $x^2 = 0$, so all other copies of $\mathbb{k}$ die.

Exercise 45. Investigate the commutative graded algebra $\mathbb{k}[x, y]$, where we set $\deg(x) = 0$ and $\deg(y) = -1$. More precisely, consider the following approaches:

- (1) Describe $\mathbb{k}[x, y]$ by hand first, as is done in the above example.
- (2) Show that if M is a graded module concentrated in degree zero (i.e. just an ordinary module) and N is a graded module, then $M \otimes_{\mathbb{k}} N$ is the graded module $(M^0 \otimes_{\mathbb{k}} N^i)_{i \in \mathbb{Z}}$.
- (3) Reason that $\mathbb{k}[x, y] \cong \mathbb{k}[x] \otimes_{\mathbb{k}} \mathbb{k}[y]$ in order to describe $\mathbb{k}[x, y]$.
- (4) Reason that $\mathbb{k}[x, y] \cong \mathbb{k}[x][y]$ in order to describe $\mathbb{k}[x, y]$.

18.5 Appendix: Modules over graded algebras (TBD)

See [Yek19, Lemma 3.1.30].

Recall that we have fixed a commutative ring $\mathbb{k}$. Now, let us fix a graded $\mathbb{k}$ -algebra A . What does it mean to have a graded A -module?

Definition 18.26. A *graded left A -module* is a graded $\mathbb{k}$ -module M together with a left action of A

$$A \otimes_{\mathbb{k}} M \rightarrow M, \quad a \otimes m \mapsto am,$$

which is unital and associative. Dually, a *graded right A -module* is a graded $\mathbb{k}$ -module M together with a *right* action of A

$$M \otimes_{\mathbb{k}} A \rightarrow M, \quad m \otimes a \mapsto ma.$$

Let M, N be graded left A -modules. A morphism $f: M \rightarrow N$ of graded $\mathbb{k}$ -modules is *A -linear* if

$$\begin{array}{ccc} A \otimes_{\mathbb{k}} M & \xrightarrow{1 \otimes f} & A \otimes_{\mathbb{k}} N \\ \downarrow & & \downarrow \\ M & \xrightarrow{f} & N \end{array}$$

commutes. The definition for graded right A -modules is similar. We obtain the categories ${}_A \mathbf{gMod}$ and $\mathbf{gMod}_A$.

Remark 18.27. Let M be a graded left A -module. Note that, by definition, we have

$$a \in A^i, m \in M^j \implies am \in M^{i+j}$$

since the action $A \otimes M \rightarrow M$ restricts to $A^i \otimes_{\mathbb{k}} M^j \rightarrow M^{i+j}$. A similar statement is true for right A -modules.

Remark 18.28. Note that the definition of A -linearity makes sense for a morphism of any degree. However, there is a perhaps odd technical detail here: if f is a graded $\mathbb{k}$ -module morphism of degree i between graded A -modules, then the concrete equations required for f to be A -linear depends on whether we are working with left or right modules. The reason is the Koszul sign rule. If we are in the case of right graded A -modules, then everything is easy since $(f \otimes 1)(x \otimes a) = f(x) \otimes a$, so we require that $f(xa) = f(x)a$. On the other hand, for *left* graded A -modules, we are working with $1 \otimes f$, and therefore we require that

$$f(ax) = (-1)^{ij} a f(x), \quad a \in A^j.$$

Remark 18.29. A graded $\mathbb{k}$ -algebra A induces a related graded $\mathbb{k}$ -algebra A^{op} , whose multiplication is “opposite” to the one in A . In particular, one applies the symmetry isomorphism to define a new algebra structure by

$$* : A \otimes_{\mathbb{k}} A \xrightarrow{\tau} A \otimes_{\mathbb{k}} A \xrightarrow{\text{mult.}} A, \quad \text{i.e.} \quad x * y = (-1)^{ij} yx, \quad x \in A^i, y \in A^j.$$

There is an equivalence of categories ${}_A \mathbf{gMod} \simeq \mathbf{gMod}_{A^{\text{op}}}$; a graded left A -module M can be endowed with a right A^{op} action induced by the morphism

$$M \otimes_{\mathbb{k}} A \xrightarrow{\tau} A \otimes_{\mathbb{k}} M \xrightarrow{\text{act.}} M.$$

Notably, we need only ever consider graded right A -modules. Observe that if A is graded commutative then $A = A^{\text{op}}$.

Remark 18.30. For a graded left A -module M , the action $A \otimes_{\mathbb{k}} M \rightarrow M$ transposes by adjunction to a morphism

$$A \rightarrow \underline{\mathbf{gMod}}_{\mathbb{k}}(M, M) =: \underline{\text{End}}_{\mathbb{k}}(M).$$

This yields an equivalent way of thinking about graded left A -modules. Similarly, a graded right A -module structure can be encoded by a map $A^{\text{op}} \rightarrow \underline{\text{End}}_{\mathbb{k}}(M)$.

What we would like to do is to extend the tensor-Hom adjunction for graded $\mathbb{k}$ -modules to the setting of graded A -modules. For this, we need to adapt both the inner $\text{Hom}_{\underline{\mathbf{gMod}}_{\mathbb{k}}}(-, -)$ and the tensor product $- \otimes_{\mathbb{k}} -$ to their respective A -linear variants. The easiest is the former, as we already know what it means for a morphism to be A -linear:

Definition 18.31. Let A be a graded $\mathbb{k}$ -algebra, and let $M, N \in \mathbf{gMod}_A$. We let

$$\underline{\mathbf{gMod}}_A(M, N)^i \subseteq \underline{\mathbf{gMod}}_{\mathbb{k}}(M, N)^i$$

consist of those morphisms of degree i which are A -linear. This defines a graded A -module $\underline{\mathbf{gMod}}_A(M, N)$, which we may also sometimes denote $\underline{\text{Hom}}_A(M, N)$.

The tensor product is a little bit trickier, but in the end, we would basically just like to take the tensor product $- \otimes_{\mathbb{k}} -$ we know about and impose some further equations on it. We may observe that if M is a right graded A -module and N is a left graded A -module, we have two maps

$$\ell, r: M \otimes_{\mathbb{k}} A \otimes_{\mathbb{k}} N \rightarrow M \otimes_{\mathbb{k}} N$$

namely

$$\ell = (\text{act.})_M \otimes 1_N, \quad \text{and} \quad r = 1_M \otimes (\text{act.})_N.$$

Now, $\mathbf{gMod}_{\mathbb{k}}$ is an Abelian category, as all functor categories with codomain an Abelian category are, so we can take the difference of these two maps. Concretely,

$$(\ell - r)(x \otimes a \otimes y) = xa \otimes y - x \otimes ay.$$

With this in mind, it makes sense to make the following definition.

Definition 18.32. Let M be a right graded A -module and let N be a left graded A -module. We define their tensor product $M \otimes_A N$ over A to be

$$M \otimes_A N := \text{coker}(M \otimes_{\mathbb{k}} A \otimes_{\mathbb{k}} N \xrightarrow{\ell - r} M \otimes_{\mathbb{k}} N).$$

We will abusively still write $x \otimes y$ for the equivalence class in the cokernel. This becomes a *graded A -bimodule* with the action

$$a(x \otimes y) = x \otimes ay = xa \otimes y = (x \otimes y)a.$$

Remark 18.33. It might help to make the grading explicit. To this end, let us investigate the tensor products of the form

$$M \otimes_{\mathbb{k}} A \otimes_{\mathbb{k}} N.$$

Arbitrarily, let us pick the bracketing $(M \otimes_{\mathbb{k}} A) \otimes_{\mathbb{k}} N$, and compute

$$\begin{aligned} ((M \otimes_{\mathbb{k}} A) \otimes_{\mathbb{k}} N)^n &= \coprod_{i+j=n} (M \otimes_{\mathbb{k}} A)^i \otimes_{\mathbb{k}} N^j \\ &= \coprod_{i+j=n} \left(\coprod_{p+q=i} M^p \otimes_{\mathbb{k}} A^q \right) \otimes_{\mathbb{k}} N^j \\ &\cong \coprod_{i+j=n} \left(\coprod_{p+q=i} M^p \otimes_{\mathbb{k}} A^q \otimes_{\mathbb{k}} N^j \right) = \coprod_{i+t+j=n} M^i \otimes_{\mathbb{k}} A^t \otimes_{\mathbb{k}} N^j. \end{aligned}$$

so that the n th graded piece of $M \otimes_{\mathbb{k}} A \otimes_{\mathbb{k}} N$ is obtained by taking the coproduct of objects

$$M^i \otimes_{\mathbb{k}} A^t \otimes_{\mathbb{k}} N^j, \quad i + t + j = n.$$

In particular, we see that

$$(M \otimes_A N)^n = \operatorname{coker} \left(\coprod_{i+t+j=n} M^i \otimes_{\mathbb{k}} A^t \otimes_{\mathbb{k}} N^j \rightarrow \coprod_{p+q=n} M^p \otimes_{\mathbb{k}} N^q \right)$$

where the map has exactly the same formula as before.

For simplicity, we will now restrict to the case where A is graded commutative. This is to avoid spending time talking about bimodules.

Proposition 18.34. *We have*

$$\mathbf{gMod}_A(M \otimes_A N, L) \cong \mathbf{gMod}_A(M, \underline{\mathbf{gMod}}_A(N, L)).$$

19 Differential graded modules & algebras (WIP)

In Lecture 18, we discussed algebra in the world of graded objects. The motivation is that if we forget the differential on a chain complex of modules, the structure you have is that of a *graded* module. In this lecture, we will reintroduce the differential to talk about differential graded modules and algebras.

19.1 Differential graded modules

As in Lecture 18, we will fix a commutative ring $\mathbb{k}$. Recall that a graded $\mathbb{k}$ -module is an object of the category

$$\mathbf{gMod}_{\mathbb{k}} = \prod_{\mathbb{Z}} \mathbf{Mod}_{\mathbb{k}}.$$

We may picture some $M \in \mathbf{gMod}_{\mathbb{k}}$ as a sequence

$$\dots \quad M^{i-1} \quad M^i \quad M^{i+1} \quad \dots$$

and this makes it fairly clear where to go next. The category $\mathbf{gMod}_{\mathbb{k}}$ is equipped with a *shift* functor

$$(1) : (M^i)_{i \in \mathbb{Z}} \mapsto (M^{i+1})_{i \in \mathbb{Z}}.$$

Composing this with itself (or the inverse) iteratively allowed us to define morphisms of degree i between graded modules. For simplicity, we now introduce some notation.

Notation 19.1. Writing “a morphism $M \rightarrow N$ of degree i ” will mean a morphism $M \rightarrow N(i)$. Given a morphism $f : M \rightarrow N$ of degree i and a morphism $g : N \rightarrow L$ of degree j , we let the composite $g \circ f$ be the morphism $M \rightarrow L$ of degree $i + j$ given by

$$M \xrightarrow{f} N(i) \xrightarrow{g(i)} L(i+j).$$

The same thing can be done for graded objects in any category $\mathcal{C}$, and we adopt the same convention there.

Remark 19.2. The above notational convention makes it clear that $\mathbf{gMod}_{\mathbb{k}}(M, M)$ is a graded $\mathbb{k}$ -algebra for any $M \in \mathbf{gMod}_{\mathbb{k}}$.

Definition 19.3. Let $\mathcal{C}$ be a category with a zero object. A *differential graded object*, or *dg-object*, in $\mathcal{C}$ is a pair (x, d) consisting of a graded object x in $\mathcal{C}$ and a morphism $d : x \rightarrow x$ of degree one such that $d^2 = 0$. A morphism $(x, d_x) \rightarrow (y, d_y)$ of dg-objects is a morphism $x \rightarrow y$ of the underlying graded objects such that

$$\begin{array}{ccc} x & \xrightarrow{d_x} & x \\ \downarrow & & \downarrow \\ y & \xrightarrow{d_y} & y \end{array}$$

commutes. Letting $\mathcal{C} = \mathbf{Mod}_{\mathbb{k}}$, we obtain the category $\mathbf{dgMod}_{\mathbb{k}}$ of *dg-modules* over $\mathbb{k}$. If the context is clear, we will omit the differential in a dg-object.

Remark 19.4. The category of dg-objects in an Abelian category $\mathcal{A}$ is either definitionally the same as, or in any case easily isomorphic to, the category of chain complexes $\mathbf{Ch}(\mathcal{A})$. In particular, $\mathbf{dgMod}_{\mathbb{k}} \cong \mathbf{Ch}(\mathbb{k})$. Thus, we can view dg-objects/dg-modules as just another way to encode the same data we are used to. As a consequence, we also see that $\mathbf{dgMod}_{\mathbb{k}}$ is a $\mathbb{k}$ -linear (i.e. enriched in $\mathbb{k}$ -modules) Abelian category admitting small products and coproducts, though this is also very easy to check independently.

Remark 19.5. We may identify $\mathbf{gMod}_{\mathbb{k}}$ with a full subcategory of $\mathbf{dgMod}_{\mathbb{k}}$, namely as those dg-modules with zero differential.

The shift (1) on $\mathbf{gMod}_{\mathbb{k}}$ induces a shift [1] on $\mathbf{dgMod}_{\mathbb{k}}$. It is given as follows:

Definition 19.6. We define the *shift functor*

$$[1] : \mathbf{dgMod}_{\mathbb{k}} \rightarrow \mathbf{dgMod}_{\mathbb{k}}, \quad (M, d_M) \mapsto (M(1), -d_M(1)).$$

That is, $d_{M[1]} = -d_M(1)$, so that $d_{M[1]}^i = -d_M^{i+1}$. As with the functor (1), this is an automorphism of $\mathbf{dgMod}_{\mathbb{k}}$, and we iteratively define [i] for any $i \in \mathbb{Z}$.

19.2 Tensor products and Hom in the differential graded world

We know how to take tensor products of graded modules, namely: apply the functor Tot to the obvious *bigraded* object. If we now have dg-modules, all we are missing is a differential.

Remark 19.7. Before jumping into defining a differential on the tensor product of two dg-modules, let us briefly discuss the Koszul sign rule and its effect on tensor products of graded modules.

Definition 19.8. Let $(M, d_M), (N, d_N) \in \mathbf{dgMod}_{\mathbb{k}}$. We define

$$(M, d_M) \otimes_{\mathbb{k}} (N, d_N) := (M \otimes_{\mathbb{k}} N, d_{M \otimes_{\mathbb{k}} N}),$$

where $d_{M \otimes_{\mathbb{k}} N} = d_M \otimes 1_N + 1_M \otimes d_N$. By the Koszul sign rule, this acts on elementary tensors by

$$d_{M \otimes_{\mathbb{k}} N}(x \otimes y) = d_M(x) \otimes y + (-1)^i x \otimes d_N(y), \quad x \in M^i.$$

Exercise 46. Verify that $d_{M \otimes_{\mathbb{k}} N}^2 = 0$.

Remark 19.9. There is a notion of *differential bigraded objects*, or more commonly *double complexes*. Just as we did with the Tot-construction from Lecture 18, one can take the total complex of a double complex and obtain a dg-object. There is an obvious way to form a "bigraded" version of the tensor product of two dg-modules, and the above-defined dg-module tensor product can be recovered through applying the total complex construction to this bigraded version.

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